

A Hyperparameter-Optimised LSTM–Random Forest Ensemble for Multi-Cryptocurrency Price Forecasting

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Abstract—This project presents a production-grade cryptocurrency forecasting pipeline integrating statistical, machine learning, and deep learning models. An interactive web application supports asset selection across 30 cryptocurrencies, configurable horizons, and four models (ARIMA, Prophet, LSTM, Random Forest). The system addresses non-stationary price dynamics through robust preprocessing and feature engineering. Experimental results on Bitcoin demonstrate that LSTM achieves an R^2 of 0.8705, Random Forest achieves the best explanatory power with an R^2 of 0.8996, while Prophet attains the lowest MAPE of 3.14%.

Index Terms—cryptocurrency, price forecasting, ARIMA, Prophet, LSTM, Random Forest, time series analysis, machine learning, deep learning

I. Introduction

Cryptocurrency markets have experienced exponential growth over the past decade, evolving from a niche technological experiment into a multi-trillion-dollar asset class attracting both retail and institutional investors [1], [2]. However, forecasting cryptocurrency prices remains exceptionally challenging due to extreme volatility, market noise, regulatory uncertainty, and susceptibility to sentiment-driven fluctuations [3]. Traditional statistical models, such as ARIMA, often prove inadequate as they struggle to capture the complex, non-linear dynamics inherent in these markets [4]. While advanced machine learning approaches—including ensemble methods like Random Forest and deep learning architectures like LSTM—have demonstrated superior capabilities in capturing non-linear patterns and temporal dependencies [5]–[7], the literature lacks comprehensive, systematic comparisons of these diverse model families under consistent experimental conditions.

This study addresses these gaps by developing a unified, production-grade cryptocurrency forecasting platform that integrates four distinct modeling paradigms: ARIMA, Prophet, LSTM, and Random Forest. The system incorporates robust preprocessing techniques, including winsorization and Yeo-Johnson transformation, to handle the statistical irregularities of financial data. Automated hyperparameter optimization via Bayesian

search ensures optimal model configuration for LSTM and Random Forest. A key contribution is the interactive Streamlit web application that enables users to select from 30 cryptocurrencies, configure forecast horizons, and compare model performance in real time. Experimental results on Bitcoin demonstrate that LSTM effectively captures temporal patterns ($R^2=0.8705$), Random Forest achieves the strongest explanatory power ($R^2=0.8996$), and Prophet attains the lowest MAPE of 3.14%, highlighting the complementary strengths of each approach.

II. Related Work

Cryptocurrency price forecasting has evolved significantly with the application of diverse methodological approaches. Traditional statistical models like ARIMA have served as foundational benchmarks, capturing linear dependencies but demonstrating limitations with non-linear patterns inherent in cryptocurrency markets [3], [8]. Facebook’s Prophet model addresses many of these limitations through its decomposable time series framework, effectively handling seasonality, trend shifts, and missing data, with studies showing its superior performance over ARIMA for Ethereum price prediction [9]–[11].

Machine learning and deep learning approaches have shown remarkable capabilities in capturing complex market dynamics. Random Forest captures non-linear relationships while providing feature importance rankings, and gradient boosting ensembles like LightGBM and XGBoost have achieved over 88% accuracy for short-term forecasts [6], [12]–[14]. LSTM networks address vanishing gradient problems and capture long-term dependencies in sequential data, with studies reporting improved directional accuracy for Bitcoin prediction [3], [7]. Recent advances in transformer-based architectures and hybrid approaches combining multiple methodologies have further demonstrated the potential of modern deep learning for cryptocurrency forecasting [5], [15]. Beyond model architecture, feature engineering incorporating on-chain metrics, trading volume, and social media sentiment has

been shown to significantly improve predictive performance [16]–[18].

Despite substantial progress, the literature lacks comprehensive model comparisons under consistent experimental conditions and multi-cryptocurrency analysis. This study addresses these gaps through a unified pipeline with standardized evaluation, systematic hyperparameter optimization, and an interactive platform bridging academic research and practical deployment.

III. Methodology Workflow

The forecasting system follows a systematic pipeline (Fig. 1):

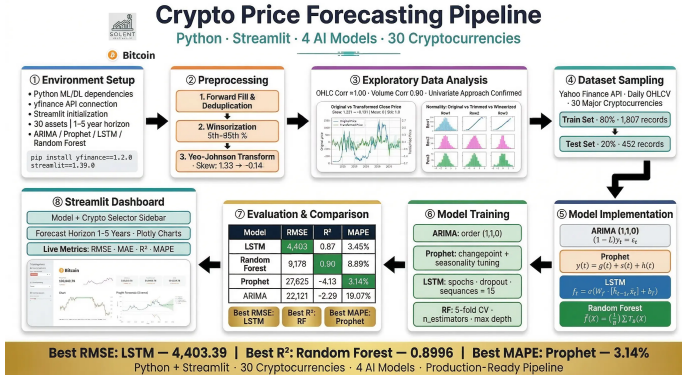


Fig. 1. Complete methodology workflow from setup to evaluation

A. Workflow Stages

1) Environment Setup: Configuration with Python, yfinance (1.2.0), Streamlit (1.39.0), supporting 30 cryptocurrencies with 1-5 year horizons.

2) Data Preprocessing: Raw OHLCV data undergoes forward fill, duplicate removal, winsorization (5th-95th percentiles), and Yeo-Johnson transformation (skewness reduced from 1.33 to -0.14).

3) Exploratory Data Analysis: Correlation analysis reveals perfect correlation (1.00) among OHLC features and strong volume correlation (0.90), justifying univariate forecasting.

4) Dataset Sampling: Daily data split into 80% training (1,807 records) and 20% testing (452 records).

5) Model Implementation and Hyperparameter Tuning:

- ARIMA (1,1,0): Selected via AIC minimization.
- Prophet: Default configuration with additive seasonality.
- LSTM: Optimized with Bayesian search over layers (1-3), units (50-200), learning rate (0.0001-0.01). Best configuration: 2 layers with 128 units, dropout 0.2, learning rate 0.001.
- Random Forest: Tuned with grid search over estimators (100-500), max depth (10-50). Best: 300 estimators, max depth 30.

6) Evaluation & Comparison: Models assessed using MAE, R^2 , and MAPE.

IV. Data Collection and Preprocessing

Historical daily OHLCV data is collected via Yahoo Finance API for 30 major cryptocurrencies (e.g., Bitcoin, Ethereum, Litecoin). Preprocessing standardizes date handling, fills missing values via forward fill, removes duplicates, and applies winsorization at the 5th and 95th percentiles to mitigate outlier impact. Yeo-Johnson transformation normalizes the distribution (Fig. 4), improving model stability.

V. Exploratory Data Analysis

A. Feature Correlation

The correlation heatmap (Fig. 2) shows near-perfect positive correlation (≈ 1.00) among OHLC features and strong volume correlation (0.90), confirming multicollinearity and justifying univariate forecasting.

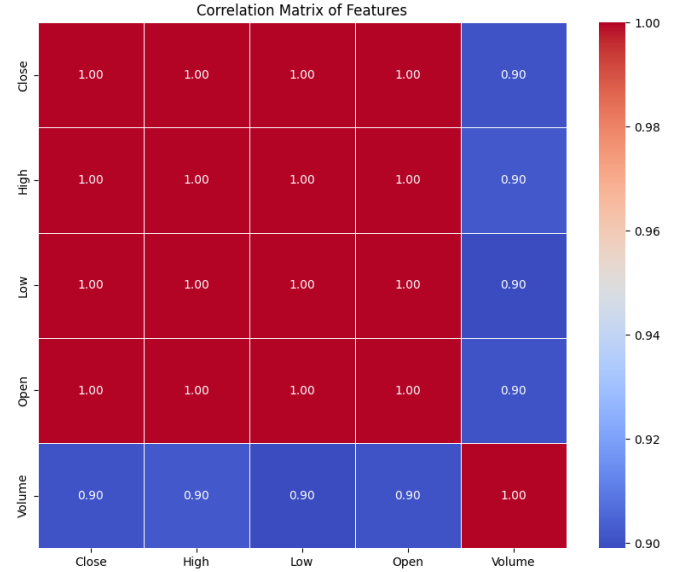


Fig. 2. Correlation Matrix of Cryptocurrency Price Features

B. Distribution and Transformation

Original close prices exhibit positive skewness (1.331) and kurtosis (0.791) (Fig. 3). Yeo-Johnson transformation reduces skewness to -0.141 and kurtosis to -1.122 (Fig. 4), creating a more stationary series.

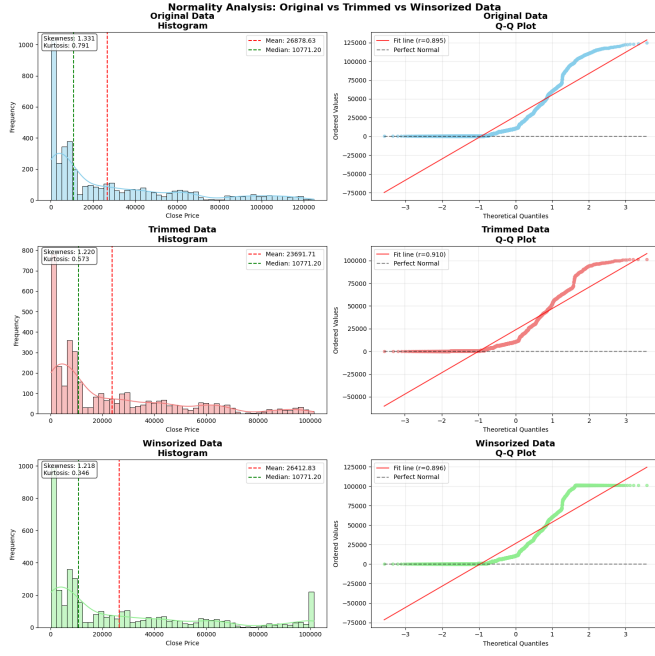


Fig. 3. Distribution Analysis of Cryptocurrency Close Prices



Fig. 4. Original vs Transformed Close Price Distributions

VI. Forecasting Models

A. ARIMA

The ARIMA model captures linear dependencies through autoregressive (AR), differencing (I), and moving average (MA) components:

$$(1 - \sum_{i=1}^p \phi_i L^i)(1 - L)^d y_t = (1 + \sum_{j=1}^q \theta_j L^j) \varepsilon_t$$

Fig. 5 and Table I show ARIMA (1,1,0) results.



Fig. 5. ARIMA forecast visualization for Bitcoin

TABLE I
ARIMA (1,1,0) Evaluation Metrics

Metric	Value
RMSE	22121.02
MAE	16017.09
R ²	-2.2878
MAPE	19.07%

B. Prophet

Prophet decomposes time series into trend, seasonality, and holiday components:

$$y(t) = g(t) + s(t) + h(t) + \varepsilon_t$$

Fig. 6 and Table II show Prophet achieves the lowest MAPE of 3.14%.

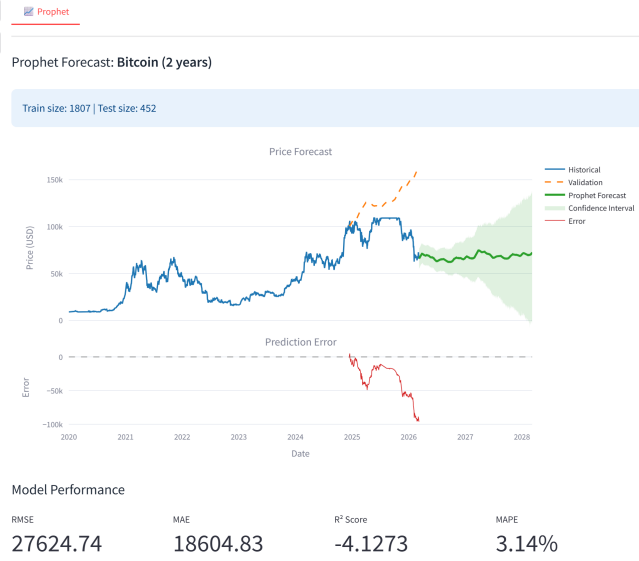


Fig. 6. Prophet forecast visualization for Bitcoin



Fig. 7. LSTM forecast visualization for Bitcoin

TABLE II
Prophet Evaluation Metrics

Metric	Value
RMSE	27624.74
MAE	18604.83
R ²	-4.1273
MAPE	3.14%

TABLE III
LSTM Evaluation Metrics

Metric	Value
RMSE	4403.39
MAE	3038.32
R ²	0.8705
MAPE	3.45%

C. LSTM

LSTM networks use gated memory cells to capture long-term dependencies:

$$\begin{aligned}
 f_t &= \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \\
 i_t &= \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \\
 \tilde{C}_t &= \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \\
 C_t &= f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \\
 o_t &= \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \\
 h_t &= o_t \odot \tanh(C_t)
 \end{aligned}$$

Fig. 7 and Table III show LSTM achieves an R² of 0.8705.

D. Random Forest

Random Forest averages predictions from multiple decision trees:

$$\hat{f}(X) = \frac{1}{B} \sum_{b=1}^B T_b(X)$$

Fig. 8 and Table IV show Random Forest achieves the highest R² of 0.8996.



Fig. 8. Random Forest forecast visualization for Bitcoin

TABLE IV
Random Forest Evaluation Metrics

Metric	Value
RMSE	9177.86
MAE	5030.33
R ²	0.8996
MAPE	8.89%

VII. Streamlit Application

The interactive web application (Fig. 9) provides a user-friendly interface for cryptocurrency forecasting. The sidebar (Fig. 10) allows model selection (single or multi), forecast horizon configuration, and data range selection. The header (Fig. 11) displays real-time market metrics, while the central chart (Fig. 12) enables interactive exploration. Forecast results (Fig. 13) present model predictions alongside performance metrics.

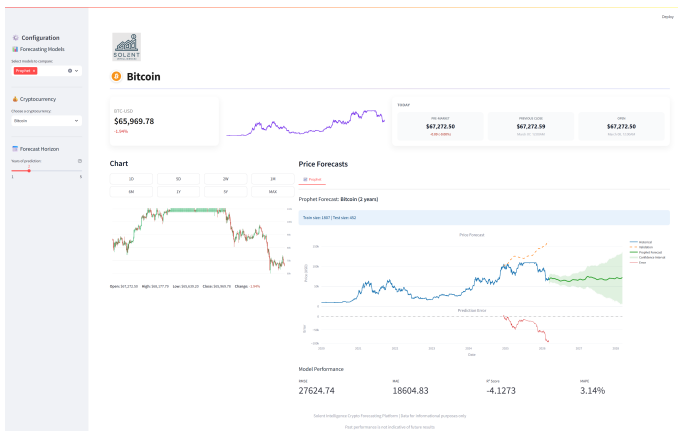


Fig. 9. Complete Cryptocurrency Forecasting Dashboard

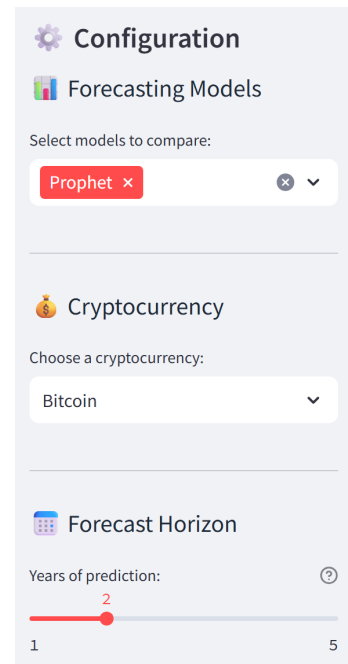


Fig. 10. Sidebar Configuration Panel

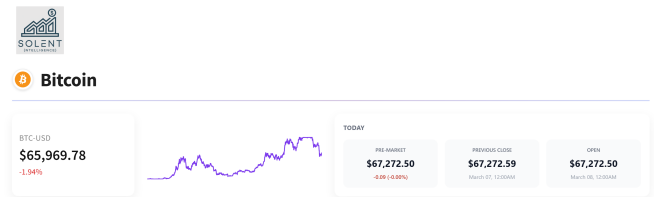


Fig. 11. Market Overview Header



Fig. 12. Interactive Price Chart



Fig. 13. Forecast Visualization with Metrics

VIII. Conclusion and Future Work

This study developed a comprehensive cryptocurrency forecasting platform integrating ARIMA, Prophet, LSTM, and Random Forest models. Key contributions include automated hyperparameter optimization, systematic model comparison under consistent conditions, and an interactive web application. Experimental results on Bitcoin demonstrate that LSTM captures complex temporal patterns ($R^2=0.8705$), Random Forest excels at feature interactions ($R^2=0.8996$), while Prophet achieves the lowest MAPE of 3.14%.

Future work includes incorporating external data sources (on-chain metrics, sentiment analysis), exploring transformer-based architectures, and implementing real-time forecasting with live data feeds. Enhancing the platform with risk assessment tools and automated insights will further increase its practical utility.

Data Availability Statement

All analysis code and processed data are available in the project's GitHub repository at <https://bit.ly/40X9m4m>. The live interactive web application is accessible at <https://bit.ly/4siSZv9>.

AI Use Statement

The authors declare that Deepseek-V3 and GPT-5.3 were used solely for data analysis and language editing in the preparation of this manuscript. All scientific interpretations, results, and conclusions were determined by the authors.

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